

Srikanth Goli

Contact Information	H.No: 2-88, Kalvalapally, Munugode Nalgonda, Telangana, India-508244. golisrikanth1989	+91-9885533518 golisrikanth1989@gmail.com golisrikanth1989
Hardware & Software Skills	• Programming: Python (Sionna (Physical Layer Library)), C/C++ and Go. • Tools: MATLAB, Mathematica, Wireshark & LaTeX. • Hardware: NI USRPs (B210 & N310), Quectel, Rasberry Pi, STM32 μcontroller.	
Research Interests	5G & 6G Open-source Tools Signal Processing (AI/ML) for Wireless	Non-Terrestrial Networks Explore Ramanujan Subspaces in Other Fields
Summary	• 7+ Yrs of experience in deploying and managing 4G/5G testbed using NI SDRs (B210 and N310) and open-source stacks Open5GS, free5GC, OAI and srsRAN. • Physical Layer Communication: Skilled in implementing advanced wireless physical layer techniques, including OFDM, Massive MIMO, Transmitter/Receiver algorithms (Precoding and Detection) and fading channels proficiently using MATLAB and Python (3GPP TS 38.211 and 38.212). • Debugging & Testing: Experienced in debugging and analyzing 4G and 5G protocols and its interfaces using wireshark and logs. • Development : 5G NR RRC procedures, Initial Synchronization (PSS and SSS) & Handover(C/C++ and Go). Developed API frameworks for Network deployment and Management using Python. Developed novel transforms to synthesise new signals for wireless. • Ability to quickly learn new tools and adapt to contribute effectively to the project/research.	
Experience	University of Luxembourg, Luxembourg. Postdoctoral Fellow - Research Associate 5G Terrestrial and Non-Terrestrial Networks. • Involved in developing and testing the Handover over Terrestrial to Non-Terrestrial networks and viceversa with simulator/emulator. • Developing and testing the Doppler compensation algorithms for Terrestrial and Non-Terrestrial networks. • SIB-19 feature implementation in OAI stack. Bloxtel, Bangalore (US Startup). Staff Engineer 5G Handover & Roaming. • Involved in developing and testing the N2, N14 and N32 based inter AMF handover/roaming with simulator. • Handover Procedures • Hands-on experience deploying and managing of Rapid Space, Rakwireless and Baicell 5G Access Points. Dolcera ITES, Hyderabad. Research Engineer	2025 - Tilldate 2024 - 2025 2021 - 2023
	4G/5G Startup Research.	

- Developed application layer for designing 5G simulator control base.
- Debugging using the related logs and Wireshark packet analysis.
- Successfully deployed 5G SA, NSA, and 4G LTE testbed setups using open-source tools/software such as OAI,srsLTE, srsRAN, free5GC, Open5GS, and MAGMA-CORE.
- Hands-on Experience in working with USRP devices like (USRP B210, and N310s).

Patent Analyst(4 Months)

Standard Essential Patent (SEP) Mappings.

- Wi-Fi patents mapping to IEEE 802.11 standards.
- Product claim chart mappings for 4G/5G patents.
- Worked on 3GPP Release 17 and 18 study and work items

Research Experience

C.R.Rao Advance Institute of Mathematics, Statistics and Computer Science (AIMSCS), Hyderabad. 2019 - 2021
Senior Research Fellow

4G Long Term Evolution (LTE) Security and Payload Analysis.

- Setting up the 4G-LTE testbed setup using open source srsLTE or Open Air Interface tool using SDRs (USRP B210).
- Developed and Integrated the Tracking Area Update procedure (Reject and Attach) 3GPP feature in the open-source code.
- LTE layer-wise analysis using Wireshark for payload extraction.

Teaching Assistance

Rajiv Gandhi University of Knowledge Technologies(RGUKT) 2012 - 2014
Shiv Nadar University(SNU) 2014 - 2018

Labs & Tutorials Handled

Digital Electronics | Embedded Systems Prof. R N Biswas.
Signal Processing | Digital Communications Prof. Vijay Kumar Chakka.

Education

Ph.D @ Shiv Nadar University 2021,
Greater Noida, Uttar Pradesh India,
Doctor of Philosophy in Electronics & Communications Engineering CGPA : **8.9**.
Specialized in *Wireless Communications and Signal Processing*

- **Thesis Topic:** *Ramanujan Subspaces and its applications to Frequency Selective Channels*
- **Description:** This work introduces a new waveforms called Ramanujan Periodic Subspace Division Multiplexing (RPSDM), designed using Periodic Ramanujan Subspaces. This waveforms are also applied to MIMO and Massive MIMO channels and compares its performance with the OFDM waveform. RPSDM demonstrates lower Peak-to-Average Power Ratio (PAPR) and better Bit Error Rate (BER) compared to OFDM. A challenge identified is that using precoding with OFDM/RPSDM can increase PAPR during transmission, irrespective of the waveform technique. To tackle this, a precoder is developed that simultaneously reduces PAPR and interference at the receiver for both OFDM and PAPR.
- **Advisor:** Prof.Vijay Kumar Chakka.

M.Tech @ Rajiv Gandhi University of Knowledge Technologies Nuzvid, Andhra Pradesh <i>Master of Technology</i> in Electronics & Communication Engineering Specialized in <i>Computational Engineering</i>	2012-14, India, CGPA:8.4.
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- **Thesis Topic:** *Vehicle Speed Detection using Motion Blur Parameters.*
 - **Description:** This project evaluates the vehicle's speed based on blur images. First images are captured with motion effects. These effects are characterized by camera models to extract the parameters like blur length and angle. Finally, these parameters are used to evaluate the speed and direction of the vehicle.
 - **Advisors:** Mr. Harsha Vardhan, M.Tech & Mr. Madhusudhan Rao, M.Tech
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B.Tech @ CVR College of Engineering, Hyderabad, Telangana <i>Bachelor of Technology</i> in Electronics & Communications Engineering	2006-10, India, 64.63 %.
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- **Project:** *Highway Speed Information with Speed-Cum Governor*
 - **Description:** This project is about vehicle speed control based on speed limits set by roadways and traffic authorities, to control the vehicle's speed based on set speed limits by authorities and this information is communicated wireless to the vehicles based on which it allow vehicles to move with speed that is set. This is also useful if any road accidents and traffic information needs to be informed to the other vehicles coming in the same direction so that drivers can take decisions to over the waiting situations.
 - **Advisor:** Mrs. Padmavathi.T, M.Tech.
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10+2 @Gauthami Junior College Nalgonda, Telangana <i>Intermediate</i> in Mathematics Physics and Chemistry	2004-06, India, 91.1 %.
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SSC @ Vaishnavi High School Nalgonda, Telangana <i>Secondary School of Education</i>	2004, India, 84.6 %.
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Projects

1. **DC Motor Speed Control & Wireless Temperature Control:**
Technical Environment: STM32L476, Z767 and STMCubeMx
Description: These two projects were developed under the supervision of Prof. R N Biswas to demonstrate it to undergraduate students as part of student coursework.

 2. **Student Information System**
Technical Environment: Java, text files as Database.
Description: This is a coursework project in postgraduate, which aids in maintaining the information related to the students. It also helps the faculty and management to update the course details, assignment of classes, examination details, and results. It also provides grades for the students automatically depending upon their scores
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Publications PUBLISHED

1. **Goli Srikanth** and S. B. Shah et al., "RPDMA: A PAPR-Aware Multiple Access Scheme," in IEEE Open Journal of Vehicular Technology, vol. 7, pp. 800-816, 2026, doi: 10.1109/OJVT.2026.3661153.

2. S. B. Shah, N. T. Ali, **Goli Srikanth**, A. Altunaiji and D. I. Oléan, "PAPR Performance Evaluation of OFDM, RPDM, and ORPDM Multicarrier Modulation Schemes," in IEEE Open Journal of the Communications Society, vol. 6, pp. 5297-5318, 2025, doi: 10.1109/OJCOMS.2025.3579725.
3. **Goli Srikanth**, V. K. Chakka and S. B. Shah, "Ramanujan Periodic Subspace Division Multiplexing," in IET Communications, vol. 13, no. 15, pp. 2296-2303, 17 9 2019, doi: 10.1049/iet-com.2018.5828
4. **Goli Srikanth**, V. K. Chakka and S. Kumar, "Convergence of MassiveMIMO Frequency Selective Channels," TENCON 2019 - 2019 IEEE Region 10 Conference (TENCON), Kochi, India, 2019, pp. 1499-1503, doi: 10.1109/TENCON.2019.8929694.
5. **Goli Srikanth** and V. K. Chakka, "Performance analysis of linear precoders and SVD in downlink MassiveMIMO Frequency selective channels," 2016 International Conference on Signal Processing and Communication (ICSC), Noida, 2016, pp. 139-144.
6. **Goli Srikanth** and V. K. Chakka, "Space-time block based linear precoders for multi user-MassiveMIMO frequency selective broadcast channels," 2017 International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), Chennai, 2017, pp. 1746-1750.

Publications
REFEREED

1. S. B. Shah, N. T. Ali, G. Srikanth, A. Altunaiji and D. I. Oléan, "PAPR Performance Evaluation of OFDM, RPDM, and ORPDM Multicarrier Modulation Schemes," in IEEE Open Journal of the Communications Society, vol. 6, pp. 5297-5318, 2025, doi:10.1109/OJCOMS.2025.3579725.
 2. B. Shah Shaik, V. K. Chakka, **Srikanth Goli** and A. S. Reddy, "Removal of narrowband interference (PLI in ECG signal) using Ramanujan periodic transform (RPT)," 2016 International Conference on Signal Processing and Communication (ICSC), Noida, 2016, pp. 233-237.
 3. B. S. Shaik, V. K. Chakka and **Srikanth Goli**, "Ramanujan and DFT mixed basis representation for removal of PLI in ECG signal," 2017 4th International Conference on Signal Processing and Integrated Networks (SPIN), Noida, 2017, pp. 509-512.
 4. Rakesh J S, Thangapandain, **Goli Srikanth** and Vijay Kumar Chakka, "A Graph Based Clustering and Preconditioning of V-MIMO Wireless Sensor Networks", (NCC) National Conference on Communications-2019.
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PAPERS IN
COMMUNICATION

- 1 **Goli Srikanth**, Vijay Kumar Chakka and K. Venu Gopalchary. "Modulation-based Precoded V-BLAST for MIMO-RPSDM channels", IEEE Wireless Communication Letters.
 - 2 **Goli Srikanth**, Vijay Kumar Chakka and K. Venu Gopalchary. "Simultaneous PAPR reduction and MUI mitigation for Massive MIMO-RPSDM using Generalized Approximate Message Passing", IEEE Wireless Communication Letters.
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PAPERS IN
PREPARATION

- 1 **Goli Srikanth** and Vijay Kumar Chakka. "Spectral Efficient Subcarrier Index Modulation using Ramanujan Periodic Subspace Division Multiplexing."
- 2 **Goli Srikanth** and Vijay Kumar Chakka. "Ramanujan Subspace Based Space-Time

Achievements & Activities

Activities

- Literature survey on “4G-LTE Protocols and its Frame structures” from 2019-2020.
- Worked as 5G-India Forum (5GIF) evaluation member for *2 months* in collaboration with LNMIIT university-Jaipur, Ericsson, Qualcomm, Samsung, and Nokia.
- IEEE Executive Member at Shiv Nadar University from 2014-16.
- As a Volunteer IEEE-National System Conference(NSC)-2015 at Shiv Nadar University.
- Participated in the All India Student Young Professionals and Women in Engineering Congress (AISYWC’16) in LNMIIT Jaipur.
- Participated in the Texas Innovative Challenging Program in 2014.
- Literature survey on 5G Wireless Communications and Signal processing.
- Literature survey on “OFDM and its Wireless Applications “in 2013.

Workshops & Conferences

- Conducted workshops on MATLAB Basics, Signal, and Image Processing for Undergraduate students at Shiv Nadar University.
- Attended a three-day workshop on “Wireless Communication using Convex Optimization” at IIT Kanpur in 2014.
- Presented papers in below listed IEEE conferences,
 - ICSC-2016, WiSPNET-2017 & TENCON-2019
- Attended IEEE Conferences are listed below,
 - NCC - 2019 & Indocrypt -2019
- Attended Open Air Interface Summer 2022 Workshop (July 12 - 13) at Orange Gardens, Paris, France.
 - Presented a demo on the 5-Fi lab-as-service product.

Achievements

- Qualified as Assistant Professor in UGC-NET, 2014 as Assistant Professor.
- Secured above **90** percentile in **GATE-(2011, 2014 & 2016)** and 97% in Andhra Pradesh **PGECE-2012**.
- Secured **80%** in Preparatory-Course (45 days) conducted by **AP-IIIT (RGUKT)-Nuzvid** in the fields of MS-office, Programming skills (Raptor, Algorithms and C-language) and Soft skills.

References

Dr. Vijay Kumar Chakka(*Professor*),
Department of Electrical Engineering,
Shiv Nadar University.

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Dr. A. Satyanarayana Reddy, *Assistant Professor*,
Department of Mathematics,
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Dr. Shaik Basheeruddin Shah, *Post Doctoral Researcher*,
Math and Computer Science,
Weizmann Institute of Science.

☎: +91-8826703498,
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Personal Profile	Date of Birth	:	10-May-1989.
	Gender	:	Male.
	Nationality	:	Indian.
	Marital status	:	Married.
	Spouse Name	:	Vennela Vootkooori.
	Languages Known	:	Telugu(<i>Native</i>), English (<i>Written & Oral</i>) & Hindi (<i>Oral</i>)

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	H.No: 2-88, Kalvalapally. <i>Vill.</i> ,	
	Munugode. <i>Man.</i> , Nalgonda. <i>Dist.</i> ,	☎: +91-7396714921.
	Telangana, India-508244.	✉: golisrikanth1989@gmail.com,

Declaration	I declare that all the information above is true to the best of my knowledge and belief.
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Date:

Place:

Srikanth Goli